

HARSHVARDHAN MAHALLE

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PROFESSIONAL SUMMARY

Mechanical Engineer, skilled in maintaining and optimizing complex mechanical systems within manufacturing and power generation industries.

EDUCATION

Masters in Advance Mechanical Engineering (Grade – 2:1)

2023

University Of Strathclyde, United Kingdom

Key Modules: Machine diagnosis and condition monitoring | Engineering Composites Polymer | Industrial Metallurgy | Gas& Steam Turbines | Boiler Thermal Hydraulics | Energy Resources & Policy | Electrical Power Systems | Energy Modelling & Monitoring | Advance manufacturing processes | Metals and alloys | Material sciences | Aerodynamic Performance

Bachelor Of Technology in Mechanical Engineering (Grade – 2:1)

2018

Symbiosis Institute of Technology, India

Key Modules: Production Technology | Thermal Engineering | Fluid Mechanics and Hydraulic Machineries | Automobile Engineering | Industrial Management | CAD/CAM | Robotics | Computational Fluid Dynamics | Mechatronics | Production Planning | Automation in Manufacturing.

TECHNICAL SKILLS

- ♦ **Software:** Siemens NX, Ansys, CAD/CAM, Catia, Solid works, Creo, AutoCAD, Microsoft Office Suite, LabVIEW, MathCAD, SAP Inventor, Recap, Probad, Crackwise, VXEelement.
- ♦ **Certifications:** Ansys Training Program, Nanotechnology Essentials, Business Administration, Robotic Process Automation.

WORK EXPERIENCE

➤ Altrad Babcock

March 2024 – Current

Static Engineer

- Supervisor for Altrad for inspection of pipework's and welds and supercritical components via remote-ops NDT and Visual inspection managing over 35 personal during SIZEWELL-B 2025 and 26 outage.
- Producing CAD models for testing and analysis under different load cases using inventor and ANSYS.
- Performed laser scan using GOscan3D hand held laser scanner and postprocessing it using VX element and extracting details to produce detailed CAD model for further analysis.
- Utilised industry-approved methods and proprietary software to conduct detailed remnant life assessment, ensuring compliance with codes such as BSEN 12952:2011 and ASME III.
- Provided actionable insights by deriving stress parameters for pressure parts and assessing creep and fatigue life (based on R5 Assessment of Structural Integrity of components at high temperatures).
- Conducted level 3 spring hanger surveys in both hot and cold conditions to evaluate spring hanger support performance in a thermal power plant to ensure proper stress distribution and system integrity.
- Performed site inspection of HRSG casing, assessing structural integrity and alignment.
- Collaborated with senior engineers to evaluate casing supports for signs of wear, misalignment, or damage.
- Developed detailed models of the casing and support structures to analyze stress distribution and thermal expansion.
- Developed a 3D model of the ConocoPhillips of boiler components using Autodesk Inventor.
- Utilised ProBad software to perform minimum thickness calculations for pressure parts using GA drawings, compared results with field reports, and compiled a comprehensive analysis report analysing whether the pressure parts are within safe operational limits, highlighting any issues or necessary actions.
- Performed Fitness-for-Service (FFS) assessments per API-579/ASME FFS-1 for a pressure vessel, evaluating thinning from inspection reports. Used ANSYS and calculations to conduct a Level 3 assessment, determining if the vessel was fit for continued service.
- Analyzing & quantifying damage mechanisms in plant components such as creep & fatigue using ANSYS.
- Project Management with proficiency to learn commercial aspects of project control.
- Root Cause Analysis and Remnant Life Prediction of a thermally fatigue damaged main steam stop valve

as per PD5500 and BS7910. To-date fatigue crack growth was demonstrated via analytical tools (Ansys/CrackWise), verified against inspection reports/records, and recommendations provided.

➤ **Stellantis Automobile**

Jan 2022 - Sep 2022

● **Mechanical Design Engineer**

- Working with American teams on parts design for the new Jeep 7-seater model, Jeep Wrangler, and the new Dodge series. Utilized advanced CAD tools, including Siemens NX and Ansys, and performed Finite Element Analysis (FEA) simulations to optimize components. Achieved reduction in part weight while ensuring safety and performance.
- Designed interior trims for Alfa Romeo, emphasizing material selection for driving comfort and safety standards.
- Lessons Learned to optimizing design processes and efficiency, reducing project timelines.

● **Project Intern**

- Collaborated with teams to ensure homologation compliance, actively engaging with procurement division, and regulatory authorities.
- Benchmarking and testing procedures, implementing FEA simulations to improve component durability and safety.
- Participated in procurement processes and attended meetings, fostering effective communication to meet regulatory requirements.

➤ **JCB Heavy loaders and excavators**

Sep 2020 – Nov 2021

Maintenance Engineer

- Conducted maintenance and repairs on heavy-duty construction machinery, including JCB excavators and Hydra and loaders.
- Functioned as a Mobile Site Visit Repair Engineer, troubleshooting mechanical and hydraulic issues under field conditions.
- Conducted regular inspections of machinery and implemented preventative maintenance procedures.
- Collaborated with specialists and senior engineers for complex repairs.
- Performed maintenance of electrical systems, such as replacing light bulbs and sockets, and repairing circuit breaker panels.

TRAINING AND CERTIFICATION

- National Thermal Power Corporation Thermal Power Plant Industrial Training
- Jindal Steel and Power Direct Reduced Iron Production Process Apprenticeship
- Sheffield Fracture Mechanics
- Safety passport
- Confined space training
- IMECHE Membership

ACHIEVEMENTS

2024: Scottish Masters Championship

2023: Scottish University Games | Inter University Games | British University Swimming League.

2019: All India University Swimming Nationals | Inter University championship.

2017: Individual Champion Swimming States | CBSE Swimming Nationals | School Games Swimming

Right to Work - https://drive.google.com/drive/folders/1bVMD_9y62fQYFunuiZSYQW_JE5xCOS?usp=sharing

Certifications- https://drive.google.com/drive/folders/17vQRUJ2drsyUti-szeqUtMbHk0cOdae4?usp=drive_link